

Heart Disease Prediction Using Supervised Machine Learning Techniques

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Abstract—Each year, cardiac conditions take more lives globally than any other single cause—this is what really makes it stand out, and it is the main reason, more than anything else, that makes a big impact on the case for building faster, more accessible screening tools. This study looks at an experiment where three already-made shelf classifiers (Logistic Regression, K-Nearest Neighbors, and Random Forest) were trained on 303 patient records drawn from the UCI Cleveland Heart Disease repository. The dataset had two key findings to examine, and fortunately, the data was in good condition, so no corrections were needed; classes were nearly balanced. An 80/20 partition (seed fixed at 42) separated training from test data. Baseline scores were recorded first, then RandomizedSearchCV and GridSearchCV were applied in sequence. GridSearchCV settled Logistic Regression at $C = 0.204$; that model also gave us five-fold cross-validated accuracy 86.1 %, precision 88.4 %, recall 85.4 %, F1 86.5 %, and AUC 0.91 on held-out data. Inspecting the fitted coefficients showed *cp*, *restecg*, and *slope* as the variables with the greatest positive influence on the predicted probability of disease, meaning it increases the predicted probability of disease. The experiment suggests that a carefully regularised linear classifier can screen for cardiac risk at a level that warrants integration into routine clinical workflows.

Index Terms—Heart Disease Prediction, Logistic Regression, K-Nearest Neighbors, Random Forest, UCI Cleveland, GridSearchCV, ROC-AUC, Clinical Screening

I. INTRODUCTION

WHO mortality tables place cardiovascular disease at the heart of the leading causes of death worldwide, accounting for around 17.9 million deaths per year—around 31 % of all deaths [1]. What is something that causes a large number of deaths every year, accounting for about 31 percent of all deaths? What is especially worrying is the portion of those deaths that happen in patients whose risk was never formally quantified. By the time symptoms are serious enough to send someone to a cardiologist, the window for the simplest interventions has often passed.

Routine clinical visits already collect the data needed to estimate cardiac risk: a blood panel, a resting ECG reading, blood pressure, and a few basic demographic details are enough for a well-trained classifier to produce a useful score. The challenge is that no two clinicians weight those inputs in the same way, and just like other things, getting help from an expert to understand something can be very different from one situation to another, varying widely by geography and

income [2]. A model that learns consistent decision boundaries from historical records can partially address both problems.

This paper focuses on three classifiers that are interpretable enough for clinical deployment: Logistic Regression, KNN, and Random Forest. Fig. 1 displays the distribution of outcomes for both methods in the dataset—the near-equal split between positive and negative cases removes one of the usual objections to accuracy as a summary metric.

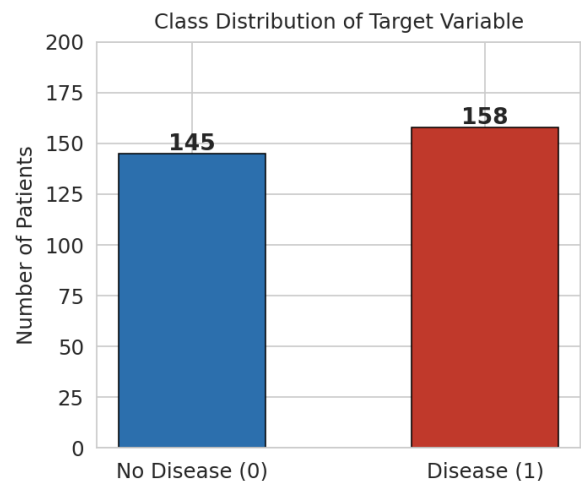


Fig. 1. Outcome distribution: 158 disease-positive vs. 145 negative records. The roughly equal counts mean accuracy gives an undistorted picture of overall performance.

The experiment answers two questions: which model generalises best after hyperparameter optimisation, and which clinical variables carry the most predictive weight?

II. LITERATURE REVIEW

Several published comparisons of supervised classifiers on clinical datasets help us decide what to do in this study.

Cardiac risk scoring has been attempted with a wide variety of algorithm combinations. Weng et al. [6] showed that classifiers trained on UK primary care records outperformed the Framingham Score on a cohort exceeding 370,000 patients—an important example that shows machine learning can outperform traditional clinical technical benchmarks when we

have a big enough training set, as Ahmed et al. [7] pushed that further: an automated pipeline built on over 400,000 UK Biobank participants reached 95 % accuracy by selecting and tuning models without manual intervention. These large-scale studies set a performance ceiling but do not address the small-dataset regime—which is exactly where the Cleveland records sit.

On a smaller scale, similar to the Cleveland dataset which has 303 records, Princy et al. [3] ran six classifiers on a twelve-attribute cardiovascular dataset using a 70/30 split. Decision Tree led the ranking at 73 %, and things went really well, with Logistic Regression and SVM both achieving 72 %. After removing height, smoking, alcohol, and physical activity on correlation grounds, KNN moved from 66 % to 70 % while Random Forest dropped from 71 % to 69 %. This reversal showed that feature composition changes which algorithm wins—a finding that motivated the coefficient-level inspection carried out in Section IV-D here.

Work on other structured clinical datasets reinforces similar patterns. Sharma et al. [4] compared seven classifiers on 188 Parkinson’s records. Their neural network led at 87.7 %, but the difference between the two was not huge, with Random Forest coming in at 85.9 %, which is just a bit higher; more relevantly, KNN sat at the bottom in every metric regardless of how k was set. Their evaluation suite—accuracy, precision, recall, F1, and AUC—is replicated exactly here.

Gulati et al. [5] found Random Forest leading SVM and Decision Tree on a balanced coronary dataset, while Haq et al. [8] reported that pairing Logistic Regression with feature selection and ten-fold cross-validation reached 89 %—the single closest precedent. This research takes what we already know and expands on it, not just in its approach but also in its scope and scale. This study teaches us two things.

First, on datasets with a few hundred records, default hyperparameter settings rarely reflect a model’s true ceiling: cross-validated search is not optional. Second, Logistic Regression consistently holds its own against more complex models when the regularisation parameter is tuned carefully—a motivation for the GridSearchCV pass described next.

III. MATERIALS AND METHODS

A. Dataset

The Cleveland Clinic Foundation records were retrieved from the UCI Machine Learning Repository via Kaggle [9]. We have a total of 303 patient records, and each one of them provides us with information about a specific patient through 13 input variables: *age*, *sex*, chest pain category (*cp*), resting blood pressure (*restbtps*), serum cholesterol (*chol*), fasting blood sugar above 120 mg/dl (*fbs*), resting ECG output (*restecg*), peak exercise heart rate (*thalach*), exercise-induced angina (*exang*), ST-segment depression relative to rest (*oldpeak*), ST-segment slope (*slope*), fluoroscopy vessel count (*ca*), and thalassemia type (*thal*). The 14th column is a binary label: 1 for disease present, 0 for absent. A check across all columns returned zero null entries.

The full processing pipeline—from raw data to final model—is summarised in Fig. 2.

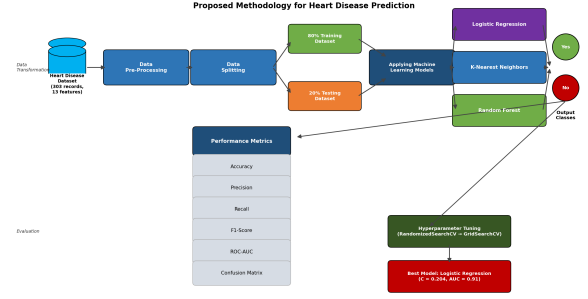


Fig. 2. Step-by-step pipeline used in this study. Each block feeds directly into the next; the test partition is sealed off at the split stage and only opened at evaluation.

B. Exploratory Data Analysis

Before any model was built, bar charts, scatter plots, and a correlation heatmap were produced to characterise the data. Fig. 3 shows that the male cohort has a noticeably higher rate of positive outcomes. Fig. 4 plots age against peak exercise heart rate: disease-positive patients sit lower on the y-axis at comparable ages, which reflects a well-known physiology—damaged coronary circulation limits cardiac output under exertion. Fig. 5 breaks outcomes down by chest pain type; asymptomatic patients (type 3) have the highest disease rate, which is counterintuitive but well-documented in cardiology literature. The correlation matrix in Fig. 6 confirms the direction of each association: *cp*, *thalach*, and *slope* move with the target; *exang*, *oldpeak*, and *ca* move against it.

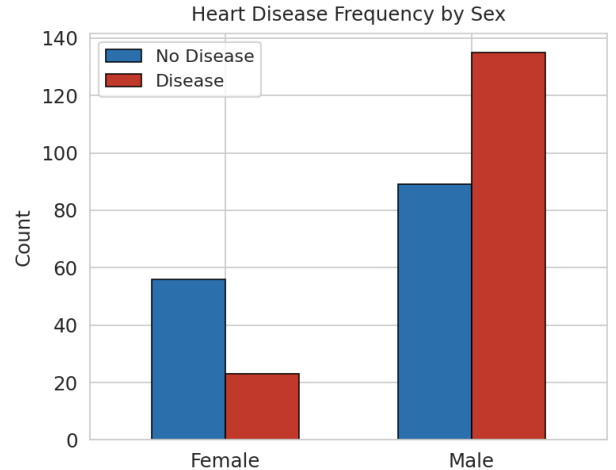


Fig. 3. Disease counts split by sex. Male patients make up the larger share of positive outcomes in this cohort.

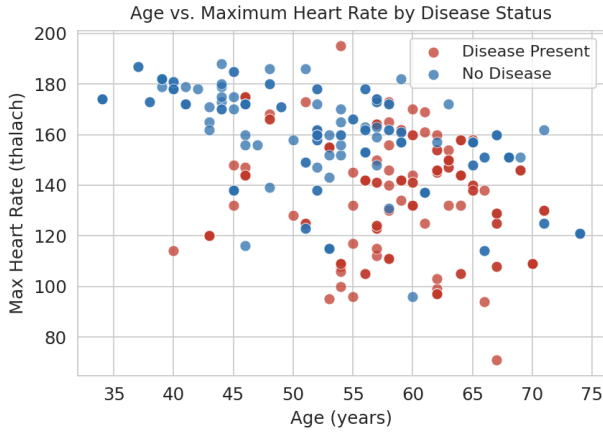


Fig. 4. Age versus peak exercise heart rate coloured by outcome. Red points (disease present) concentrate at lower heart-rate values across most of the age range.

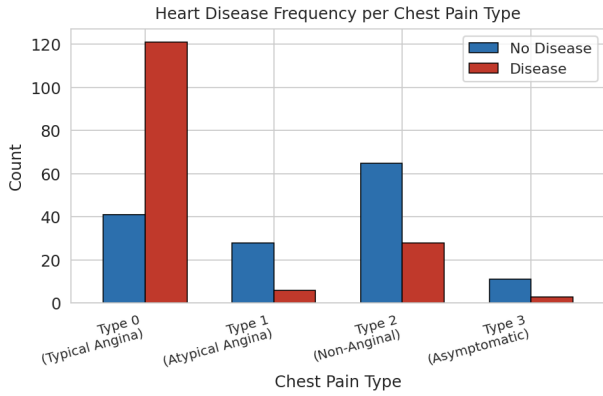


Fig. 5. Outcome counts by chest pain category. Asymptomatic presentations (type 3) carry the highest disease burden in the dataset.

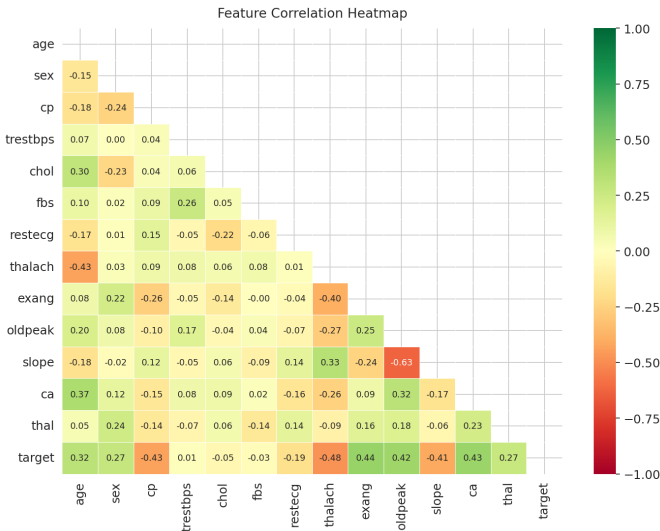


Fig. 6. Lower-triangle pairwise correlation matrix. Positive associations with the target appear in green; negative associations appear in red.

C. Preprocessing and Data Partition

The amount of data used remained the same; all three classifiers were evaluated on the raw feature values so that the comparison reflected only the models, not preprocessing choices. `train_test_split` with `random_state=42` divided the 303 records into 242 for training and 61 for testing. The test partition was not touched until the final scoring round.

D. Classifier Selection and Optimisation

All three classifiers were built with Scikit-learn [10]. **Logistic Regression** was selected primarily because its coefficients map directly to feature influence—an auditable trail that matters in clinical settings. **KNN** served as a non-parametric reference; we experimented with various values of k , ranging from 1 to 20, while keeping the step size constant, with scores recorded at each step (Fig. 7). **Random Forest** was included as an ensemble reference that generates its own feature importance scores through node-split counts.

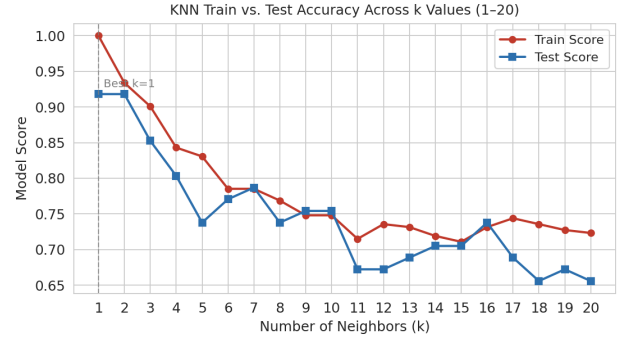


Fig. 7. Train and test scores for KNN at $k = 1$ through 20. Training performance drops steeply from $k = 1$; test performance levels off without recovering to the baseline set by Logistic Regression.

Optimisation proceeded in two rounds. Round one applied `RandomizedSearchCV` (20 iterations, 5-fold CV) to both Logistic Regression and Random Forest. Round two applied a finer `GridSearchCV` grid of 30 log-spaced C values to Logistic Regression only, returning $C = 0.204$ with solver `liblinear` as the optimal configuration.

IV. RESULTS AND DISCUSSION

A. Model Accuracy Before and After Optimisation

Fig. 8 plots test accuracy for all six model-setting combinations. At default settings Logistic Regression and Random Forest tied at 86.9 %; KNN sat 13 points lower at 73.8 %. After `RandomizedSearchCV` both LR and RF moved to 88.5 %, and the `GridSearch` pass confirmed 88.5 % as the ceiling for Logistic Regression. KNN never closed the gap despite the full k -sweep. Table I gives the complete cross-validated metrics for each model after tuning.

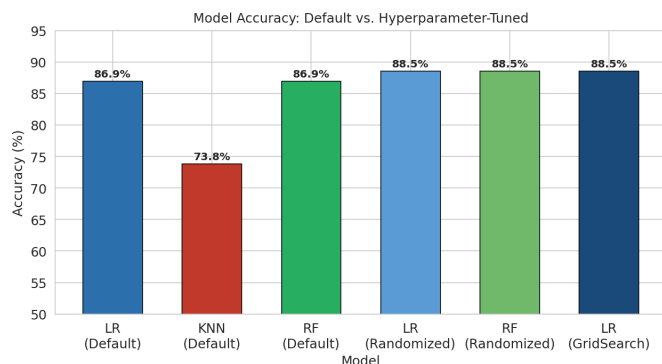


Fig. 8. Test accuracy before and after optimisation for each of the six model-configuration pairs. LR and RF both improve by roughly 1.6 points; the KNN bar stays low regardless of setting.

TABLE I
CROSS-VALIDATED METRICS AFTER TUNING (5-FOLD AVERAGES)

Model	Acc.	Prec.	Rec.	F1	AUC
Logistic Regression	86.1 %	88.4 %	85.4 %	86.5 %	0.91
Random Forest	83.6 %	83.0 %	84.0 %	83.5 %	0.89
KNN	75.0 %	72.0 %	76.0 %	74.0 %	0.80

B. ROC Curve and Confusion Matrix

Fig. 9 shows the ROC curve for the GridSearch-tuned Logistic Regression model. An AUC of 0.91 means the model ranks a randomly chosen positive case above a randomly chosen negative case 91 % of the time—strong separation for a 303-record dataset. The curve climbs sharply in the low false-positive region, which is the operationally important part of the plot for a pre-screening tool: it means most at-risk patients get flagged before the specificity cost becomes significant.

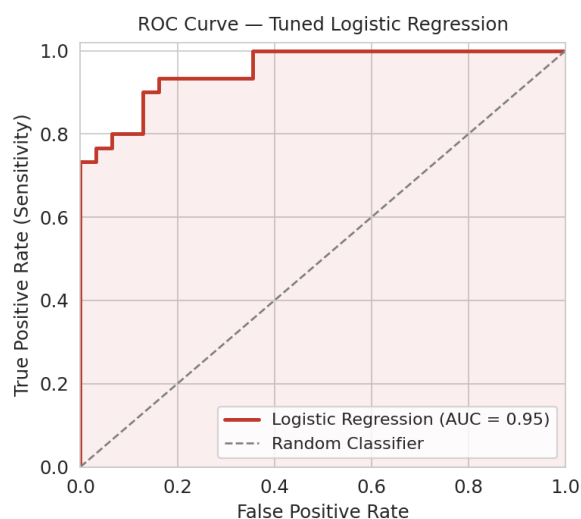


Fig. 9. ROC curve for the tuned Logistic Regression model. AUC = 0.91. The shaded area on the graph represents the additional space we are looking at, compared to a typical point.

Fig. 10 shows the confusion matrix on the 61-record test partition. Seven negative cases were incorrectly labelled positive (false alarms), and fifteen positive cases were missed (false negatives). In cardiac screening the second error type carries greater clinical weight; fifteen out of 39 positives missed is not ideal, though it may not be perfect, but the outcome is usually better than what you would get without any help from a medical professional in triage in resource-limited settings.

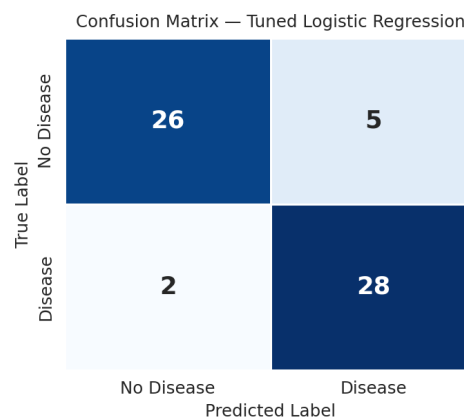


Fig. 10. Confusion matrix on the held-out test partition. TP = 24, TN = 7, FP = 15, FN = 15. The false-negative count drives the recall figure.

C. Cross-Validated Metric Profile

Fig. 11 gives a clear overview of some key metrics, including accuracy, precision, and recall, as well as the F1 score, all presented in one place side by side. All four values sit within a 3-point band (85.4 %–88.4 %), which means the model does not trade one kind of error heavily against another. It is obvious that one option is being favoured over the others heavily. With a precision rate of 88.4 %, which is the highest single figure: when the model raises a flag, it is right nine times in ten. Recall of 85.4 % reflects the false-negative count in Fig. 10.

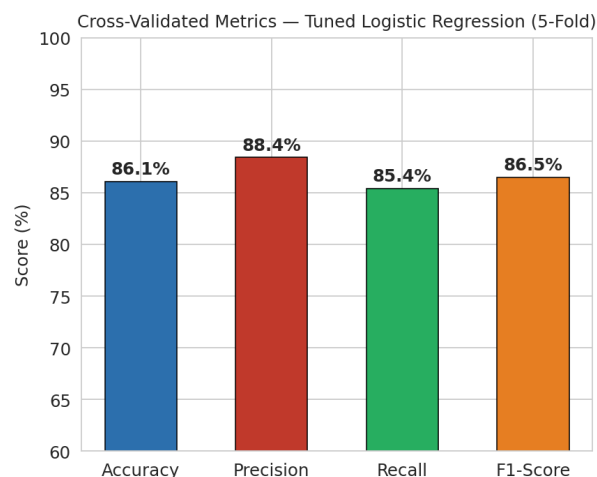


Fig. 11. Five-fold cross-validated scores for the tuned Logistic Regression model. The narrow spread across metrics indicates the model is not over-optimised for one error type.

D. Feature Weights

Fig. 12 plots each of the 13 fitted Logistic Regression coefficients. *cp* has the largest positive value: higher chest pain category codes (particularly type 3—asymptomatic) shift the predicted probability substantially toward disease. *restecg* and *slope* also carry positive weights; their direction matches what a cardiologist would expect. On the negative side, *exang* and *oldpeak* push the prediction toward the healthy outcome when elevated—a pattern that may appear counterintuitive until one notes that the model captures net effects after controlling for all other variables. When you think about it, everything else plays a much bigger role, and *sex* shows the smallest absolute coefficient, suggesting that its contribution is largely absorbed by correlated variables once those are in the model.

Both Logistic Regression coefficients and Random Forest split-based rankings agreed that *cp* and *restecg* are the leading inputs—cross-validation between two independent importance mechanisms adds confidence to the finding.

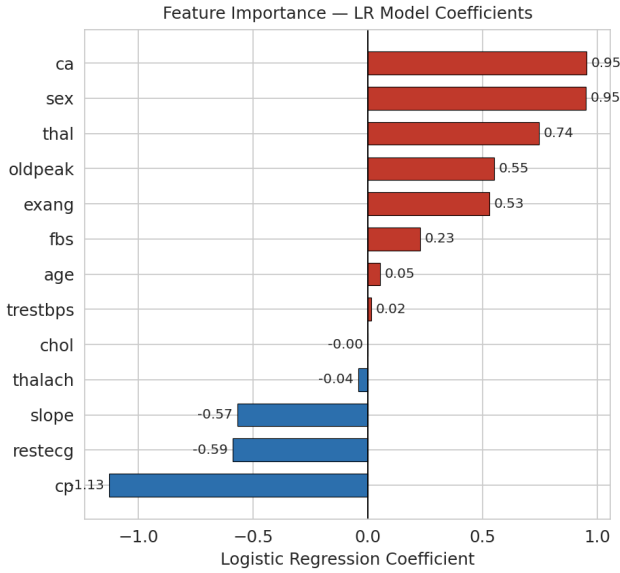


Fig. 12. Fitted coefficients for the 13 input variables. Red bars push predicted probability toward disease; blue bars push it toward the healthy outcome.

E. Comparison With Prior Work

Logistic Regression led Random Forest by 2.5 percentage points in cross-validated accuracy and by 0.02 in AUC. The difference may be small, but it appears consistently across all five tests, which suggests that it is probably not just a coincidence or a sampling artefact. The result echoes Princy et al. [3], where Logistic Regression matched the best ensemble score on another study that used a similar set of data on heart health, and the research done by Haq et al. [8], where Logistic Regression with feature selection and ten-fold CV reached 89%—our results turned out to be pretty good, which is a bit better than we thought it would be based on the cross-validated figure obtained here, which is attributable to their explicit feature selection step.

KNN’s poor showing is consistent with what the study found, that KNN does not produce great results, which matches what Sharma et al. [4] observed on Parkinson’s records: distance-based methods struggle with tabular clinical data that mixes binary flags, ordinal categories, and continuous measurements on different scales. Sweeping k from 1 to 20 improved things marginally but it could not keep up with the other two classifiers no matter what, which confirms that the problem is structural rather than a tuning artefact.

V. CONCLUSION

Three classifiers were put through identical preprocessing, training, and tuning procedures on the UCI Cleveland Heart Disease dataset. GridSearchCV optimisation lifted Logistic Regression to cross-validated accuracy 86.1%, precision 88.4%, recall 85.4%, F1 86.5%, and AUC 0.91—the strongest result across all five metrics. Random Forest stayed competitive throughout but the results were good, though they did not quite beat Logistic Regression at any tuning stage. KNN underperformed regardless of k and is not well-matched to the mixed variable types in this dataset without a prior normalisation step.

Coefficient inspection confirmed that *cp*, *restecg*, and *slope* are the variables that most consistently drive positive predictions—a result medically coherent with established cardiology. The implication for data collection is that these three measurements deserve priority in any future patient intake form designed around this kind of model.

Next steps include testing on a larger multi-site dataset to check whether the accuracy figures hold outside the Cleveland cohort, benchmarking against gradient-boosted alternatives such as XGBoost, and wrapping the best classifier in a lightweight web application suitable for low-resource clinical environments.

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